

Questions

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Q1 - 2024 (04 Apr Shift 1)

In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to :

- A. pure inductor.
- B. pure capacitor.
- C. pure resistor.
- D. combination of an inductor and capacitor.

Choose the correct answer from the options given below :

- (1) A, B and C only
- (2) A and B only
- (3) B, C and D only
- (4) A, B and D only

Q2 - 2024 (04 Apr Shift 1)

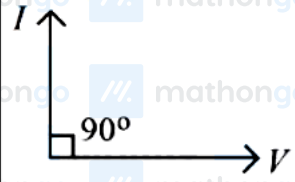
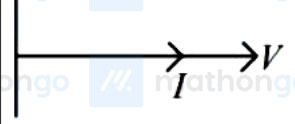
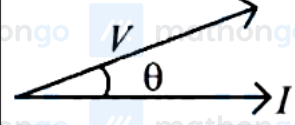
A alternating current at any instant is given by $i = [6 + \sqrt{56} \sin(100\pi t + \pi/3)]$ A. The *rms* value of the current is _____ A.

Q3 - 2024 (04 Apr Shift 2)

Questions

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Match List I with List II

LIST I		LIST II	
A.	Purely capacitive circuit	I.	
B.	Purely inductive circuit	II.	
C.	LCR series at resonance	III.	
D.	LCR series circuit	IV.	

Choose the correct answer from the options given below:

(1) A-I, B-IV, C-II, D-III

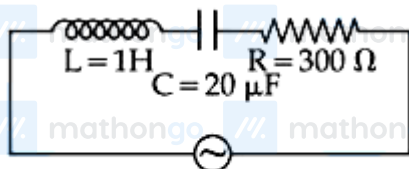
(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-I, B-IV, C-III, D-II

Q4 - 2024 (05 Apr Shift 1)

An ac source is connected in given series LCR circuit. The rms potential difference across the capacitor of $20\mu\text{F}$ is _____ V.



$$V = 50\sqrt{2} \sin 100t \text{ volt}$$

Q5 - 2024 (05 Apr Shift 2)

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A series LCR circuit is subjected to an ac signal of 200 V, 50 Hz. If the voltage across the inductor ($L = 10\text{mH}$) is 31.4 V, then the current in this circuit is _____.

- (1) 68 A
- (2) 63 A
- (3) 10 A
- (4) 10 mA

Q6 - 2024 (06 Apr Shift 1)

Given below are two statements:

Statement I: In an LCR series circuit, current is maximum at resonance.

Statement II: Current in a purely resistive circuit can never be less than that in a series LCR circuit when connected to same voltage source.

In the light of the above statements, choose the correct from the options given below:

- (1) Statement I is false but Statement II is true
- (2) Statement I is true but Statement II is false
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

Q7 - 2024 (06 Apr Shift 1)

When a *dc* voltage of 100 V is applied to an inductor, a *dc* current of 5 A flows through it. When an ac voltage of 200 V peak value is connected to inductor, its inductive reactance is found to be $20\sqrt{3}\Omega$. The power dissipated in the circuit is _____ W.

Q8 - 2024 (06 Apr Shift 2)

For a given series LCR circuit it is found that maximum current is drawn when value of variable capacitance is 25nF. If resistance of 200Ω and 100mH inductor is being used in the given circuit. The frequency of ac source is _____ $\times 10^3$ Hz. (given $\pi^2 = 10$)

Questions

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Q9 - 2024 (08 Apr Shift 1)

A LCR circuit is at resonance for a capacitor C , inductance L and resistance R . Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now:

- (1) Zero
- (2) same
- (3) halved
- (4) double

Q10 - 2024 (08 Apr Shift 2)

A coil of negligible resistance is connected in series with 90Ω resistor across 120 V , 60 Hz supply. A voltmeter reads 36 V across resistance. Inductance of the coil is :

- (1) 0.286H
- (2) 0.76H
- (3) 2.86H
- (4) 0.91H

Q11 - 2024 (08 Apr Shift 2)

An alternating emf $E = 110\sqrt{2} \sin 100t$ volt is applied to a capacitor of $2\mu\text{F}$, the rms value of current in the circuit is _____ mA,

Q12 - 2024 (09 Apr Shift 2)

A capacitor of reactance $4\sqrt{3}\Omega$ and a resistor of resistance 4Ω are connected in series with an ac source of peak value $8\sqrt{2}\text{ V}$. The power dissipation in the circuit is _____ W.

Questions

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Answer Key

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Q1 (4) mathongo Q2 (8) mathongo Q3 (1) mathongo Q4 (50) mathongo

Q5 (3) mathongo Q6 (3) mathongo Q7 (250) mathongo Q8 (10) mathongo

Q9 (4) mathongo Q10 (2) mathongo Q11 (22) mathongo Q12 (4) mathongo

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Solutions

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Q1

This is possible when phase difference is $\frac{\pi}{2}$ between current and voltage so correct answer will be (4)

Q2

$$I_{Dns} = \sqrt{\frac{\int i^2 dt}{\int dt}}$$

$$I_{Dns} = \sqrt{(6)^2 + \frac{(\sqrt{56})^2}{2}}$$

$$= \sqrt{36 + 28}$$

$$= \sqrt{64}$$

$$= 8 \text{ A}$$

Q3

A - V lags by 90° from I hence option (I) is correct.

B - V lead by 90° from I hence option (IV) is correct

C - In LCR resonance $X_L = X_C$. Hence circuit is purely resistive so option (II) is correct

D - In LCR series V is at some angle from I hence (III) is correct.

Q4

$$X_L = \omega L = 100 \times 1 = 100 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 20 \times 10^{-6}} = 500 \Omega$$

$$Z = \sqrt{(X_L - X_C)^2 + R^2}$$

$$\sqrt{(100 - 500)^2 + 300^2}$$

$$Z = 500 \Omega$$

$$i_{rms} = \frac{V_{rms}}{Z} = \frac{50}{500} = 0.1 \text{ A}$$

rms voltage across capacitor

$$V_{rms} = X_C i_{rms}$$

$$= 500 \times 0.1 = 50 \text{ V}$$

Q5

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Solutions

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Voltage across inductor $V_L = IX_L$

$$31.4 = I[L\omega]$$

$$31.4 = I[L(2\pi f)]$$

$$31.4 = I[10 \times 10^{-3}(2 \times 3.14) \times 50]$$

$$\Rightarrow I = 10 \text{ A}$$

Q6

Statement-I

$$I_m = \frac{V_m}{\sqrt{R^2 + (X_L - X_C)^2}} \text{ at resonance } X_L = X_C$$

$$\text{Thus, } I_m = \frac{V_m}{R}$$

$\therefore$ Impedence is minimum therefore I is maximum at resonance.

Statement-II

$$I = \left(\frac{V}{R} \right) \text{ in purely resistive circuit.}$$

Q7

For DC voltage

$$R = \frac{V}{I} = \frac{100}{5} = 20\Omega$$

for AC voltage

$$X_L = 20\sqrt{3}\Omega$$

$$R = 20\Omega$$

$$Z = \sqrt{X_L^2 + R^2} = \sqrt{3 \times 400 + 400} = 40\Omega$$

$$\text{Power} = i_{\text{rms}}^2 R$$

$$= \left(\frac{V_{\text{rms}}}{Z} \right)^2 \times R = \left(\frac{\frac{200}{\sqrt{2}}}{40} \right)^2 \times 20 = 250 \text{ W}$$

Q8

Solutions

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$$f_0 = \frac{1}{2\pi\sqrt{L \times C}}$$

$$f_0 = \frac{1}{2\pi\sqrt{100 \times 10^{-3} \times 25 \times 10^{-9}}}$$

$$= \frac{1}{2\pi\sqrt{25 \times 10^{-10}}}$$

$$= \frac{1}{2\pi \times 5} \times 10^5 \times \sqrt{10} \text{ Hz}$$

$$= \frac{100}{10} \times 10^3 \text{ Hz}$$

$$f_0 = 10 \times 10^3 \text{ Hz}$$

Q9

In resonance $Z = R$

$$I = \frac{V}{R}$$

$R \rightarrow$ halved

$$\Rightarrow I \rightarrow 2I$$

I becomes doubled.

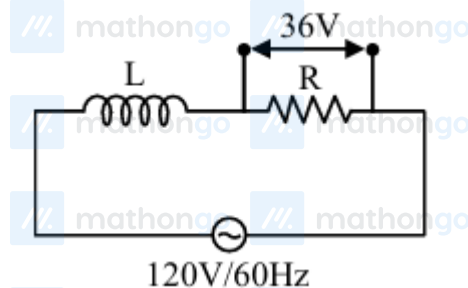
Q10

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Solutions

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$$36 = I_{\text{rms}} R$$

$$36 = \frac{120}{\sqrt{X_L^2 + R^2}} \times R$$

$$R = 90\Omega \Rightarrow 36 = \frac{120 \times 90}{\sqrt{X_L^2 + 90^2}}$$

$$\sqrt{X_L^2 + 90^2} = 300$$

$$X_L^2 = 81900$$

$$X_L = 286.18$$

$$\omega L = 286.18$$

$$L = \frac{286.18}{376.8}$$

$$L = 0.76\text{H}$$

Q11

$$C = 2\mu\text{f}; \quad E = 110\sqrt{2}\sin(100t)$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 2 \times 10^{-6}}$$

$$= \frac{10000}{2} = 5000\Omega$$

$$i_o = \frac{110\sqrt{2}}{5000}$$

$$i_{\text{rms}} = \frac{110\sqrt{2}}{5000\sqrt{2}}$$

$$= \frac{110}{5} \text{ mA}$$

$$= 22 \text{ mA}$$

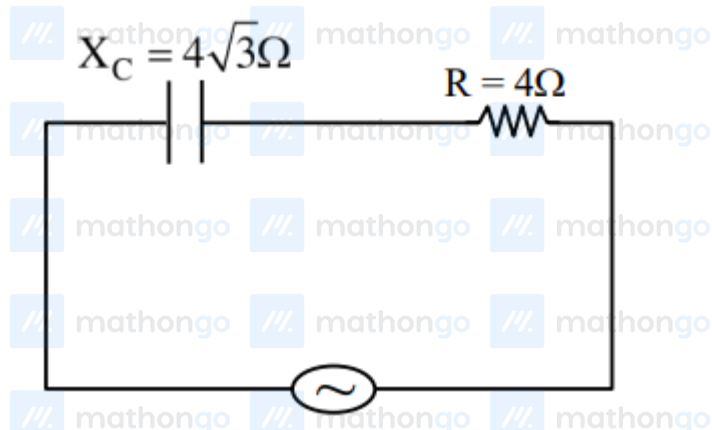
Q12

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Solutions

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$$Z = \sqrt{R^2 + X^2}$$

$$Z = \sqrt{4^2 + (4\sqrt{3})^2} = 8\Omega$$

$$V_{\text{rms}} = \frac{V}{\sqrt{2}} = \frac{8\sqrt{2}}{\sqrt{2}} = (8 \text{ V})$$

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{8}{8} = 1 \text{ A}$$

$$\text{Power dissipated} = I_{\text{rms}}^2 \times R = 1 \times 4 = (4 \text{ W})$$

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